

NORMAL APPROXIMATION TO DISTRIBUTION OF $\bar{X}$

$$\bar{X}_i \sim \text{pop}(\mu, \sigma^2)$$

Large: Population: $\mu, \sigma^2; \{X_1, \dots, X_n\}$

Small: Sample: $\bar{X}, \hat{\sigma}^2; \{\bar{X}_1, \dots, \bar{X}_n\}$

$\bar{X} \sim \text{normal}(\mu, \sigma^2/n)$ by CLT!

=> Sample Statistics

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n \bar{X}_i$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\bar{X}_i - \bar{X})^2$$

$$E(\bar{X}) = \mu$$

$$\text{VAR}(\bar{X}) = \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1}\right) \approx \sigma^2/n$$

